LESSON 142 (2.0 GROUND)

Lesson Objective

During this lesson, the instructor will evaluate the student's aeronautical knowledge on the listed areas of operation based on the current FAA Private Pilot Airman Certification Standards.

Briefing

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

References

- Aeronautical Information Manual (AIM).
    - Chapter 01 - Air Navigation.
    - Chapter 02 - Aeronautical Lighting and Other Airport Visual Aids.
    - Chapter 03 - Airspace.
    - Chapter 04 - Air Traffic Control.
    - Chapter 06 - Emergency Procedures.
- Airplane Flying Handbook. (FAA-H-8083-3C)
    - Chapter 11 - Night Operations.
    - Chapter 18 - Emergency Procedures.
- FIT AVIATION LLC – VFR Cross-Country Flight Planning Guide
- Pilot's Handbook of Aeronautical Knowledge. (FAA-H-8083-25B)
    - Chapter 05 - Aerodynamics of Flight.
    - Chapter 07 - Aircraft Systems.
    - Chapter 08 - Flight Instruments.
    - Chapter 10 - Weight and Balance.
    - Chapter 11 - Aircraft Performance.
    - Chapter 12 - Weather Theory.
    - Chapter 13 - Aviation Weather Services.
    - Chapter 14 - Airport Operations.
    - Chapter 15 - Airspace.
    - Chapter 16 - Navigation.
    - Chapter 17 - Aeromedical Factors
- Warrior Cadet/III Pilot's Information Manual.
    - Section 03 - Emergency Procedures.
    - Section 05 - Performance.
    - Section 06 - Weight and Balance.
    - Section 07 - Systems.

<u>Lesson Content</u>

<u>This is only a 2 HOUR ground lesson. This lesson plan contains a lot of information. PICK A FEW QUESTIONS FROM EACH LINE ITEM and review with the student. DO NOT ASK EVERY QUESTION ON THIS LESSON PLAN.</u>

Private Pilot Airman Certification Standards

- Go through the private pilot ACS (Airman Certification Standards).
- Ensure that the student is familiar with the applicable standards.

Certificates and Documents

- Medical certificates - 61.23.
  - Review medical classes and duration.

Under 40 Years Old			Above 40 Years Old	
12 Calendar Months ATPL/CPL Privileges	48 Calendar Months 3rd class	1st class	First 6 Calendar Months ATPL, following 6 Calendar Months, CPL	12 Calendar Months 3rd class
12 Calendar Months CPL Privileges	48 Calendar Months 3rd class	2nd class	12 Calendar Months CPL Privileges	12 Calendar Months 3rd class
60 Calendar Months - PPL Privileges		3rd class	24 Calendar Months - PPL Privileges	

  - Denial of a medical.
    - An applicant who was denied a medical has 30 days to appeal the denial in writing to the FAA. After 30 days, the applicant is considered to have withdrawn a right to a medical certificate.
    - If appealed, the medical is not considered denied until it is denied by a Federal Air Surgeon.
    - If they are found to qualify for a medical, they are issued a Special Issuance medical.
  - Review special issuances and statements of demonstrated abilities.
    - Special issuance:
      - Issued to those who have a medical condition that could get worse over time. This is issued by a Flight Surgeon.
    - Statement of demonstrated abilities:
      - Issued to those who have a medical condition that is stable and cannot worsen. This is a onetime test proving the applicant's ability to fly the aircraft. This is issued by a Flight Surgeon.
  - Review BasicMed.
    - Pilot requirements:
      - Hold a valid US driver's license.

- Must have previously held a valid FAA medical certificate, regular or special issuance, on or after July 15, 2006.
  - Cannot have been revoked, suspended, or withdrawn.
- Get a physical examination with a licensed physician every 4 years.
- Complete a BasicMed medical education course every 24 calendar months.

■ Aircraft requirements:
- Certificated to carry no more than 6 occupants.
- Has a maximum certificated takeoff weight of no more than 6,000 pounds.
  - No limitations on number of engines, gear type, or horsepower.

■ Operating requirements:
- Cannot carry more than 5 passengers.
- Cannot be operated at more than 18,000ft.
- Speed limit is 250 knots.
- Cannot fly for compensation or hire.

Cannot fly outside of the United States unless otherwise authorized.

- Private pilot certification - 61.102.
  - Eligibility requirements - 61.103.
    - 17 years of age (16 glider or balloon).
    - Able to read, speak, write, and understand English.
    - Must hold a student, sport, or recreational pilot certificate.
    - Must hold a 3rd class medical.
    - Must pass required knowledge test after:
      - Being within 24 months of age requirement for certificate.
      - Receiving a logbook endorsement from an instructor who conducted the training or reviewed a home study course on the required aeronautical knowledge areas in 61.105.
      - Obtaining an Aeronautical Knowledge Test 61.35(a)(1) + 61.103(d) + 61.105 endorsement.
    - Must receive flight training on the areas of operation in 61.107.
    - Must meet aeronautical experience requirements of 61.109.
    - Must obtain an endorsement after demonstrating satisfactory skills and knowledge to pass the required practical exam.
      - From an instructor who conducted the training in the areas of operation and certified that the person is prepared to pass the required practical test.
      - Endorsement for Flight Proficiency/Practical Test 61.103(f) + 61.107(b) + 61.109 (valid 60 days).
    - Must pass required practical test (for practical endorsement, you need flight proficiency and prerequisite (3 hours of training within 2 calendar months).

- Endorsements required for check ride/end-of-course.
    - Aeronautical knowledge test - 61.35(a), 61.103(d), and 61.105.
    - Flight proficiency - 61.103(f), 61.107(b), and 61.109.
    - Prerequisite for practical test - 61.39(a)(6)(iii).
- Privileges and limitations - 61.113.
    - Privileges of PPL:
        - Fly in furtherance of a business.
        - Be reimbursed for search and location expenses.
        - Act as PIC in accordance with 91.146.
        - Demo aircraft to buyers if pilot has 200TT.
        - Act as PIC towing a glider is in accordance with 61.69.
        - May conduct production flight tests in a light sport aircraft.
    - Limitation of PPL:
        - Act as PIC carrying persons or property for compensation or hire.
        - Fly for compensation or hire.
        - Pay less than the pro rata share of the operation expenses of the aircraft.

- Required logbook entries - 61.51.

Airworthiness Requirements
- Define airworthiness.
    - The aircraft must meet the type design, have all required inspections, comply with all airworthiness directives and be in a safe condition to fly.
    - Supplemental Type Certificate.
    - Type Certificate Data Sheet.
        - Ensure the student knows where to find it.
            - faa.gov/aircraft → technical information → TCDS → search your aircraft.
    - Required Inspections (AVIATES) - 91.171, 91.207, 91.409, 91.411, 91.413:
        - **A**nnual. 409
            - Every 12 calendar months.
        - **V**OR. 171
            - Every 30 days for IFR operations.
        - **1**00 hours. 409
            - Can overfly by 10 hours to get to maintenance.
                - Overflying takes away time from the next 100 hour.
            - Only required with for-hire operations.
        - **A**ltimeter. 411
            - Every 24 calendar months.
            - For IFR operations.
        - **T**ransponder. 413
            - Every 24 calendar months.

            - **E**LT. 207
                - Every 12 calendar months.
            - **S**tatic system. 411
                - Every 24 calendar months.
                - For IFR operations.
    - Airworthiness directives:
        - Define what an airworthiness directive is.
        - Three types:
            - Notice of proposed rulemaking.
            - Final rule.
            - Emergency.
        - Two methods of compliance:
            - One-time.
            - Recurring.
    - Service bulletins:
        - This is similar to an AD but from the manufacturer.
- Required documents and equipment for VFR flight.
    - Applicable regulations: 91.203 and 91.205.
    - Required documents (ARROWES):
        - **A**irworthiness certificate.
            - Does not expire but invalid if not airworthy.
            - Check tail number and signature during pre-flight.
        - **R**egistration certificate.
            - Typically expires every 7 years.
            - Check tail number and expiration date during pre-flight.
            - Different ways to lose the registration certificate:
                - Death of owner.
                - Transfer of ownership.
                - The aircraft is found doing illegal activities.
                - The aircraft is totaled.
                - The certificate holder requests in writing to cancel their registration.
                - The certificate holder loses US Citizenship.
        - **R**adio station license / Radio operator permit.
            - Obtained through the Federal Communications Commission (FCC).
            - Radio station license is for the airplane.
                - Expires every 10 years.
            - Radio operator permit is for the pilot.
                - Does not expire.
        - **O**perational limitations.
            - POH section 02.

        - **W**eight and balance.
            - POH section 06.
        - **E**quipment list.
            - POH section 02.
            - Difference between MEL and KOEL.
        - **S**upplements (FLICES):
            - **F**AA form 337.
                - Major alterations or repairs.
                - Examples of major alterations can be found in part 43.
                - Must have a STC if the aircraft has a form 337
            - **L**ife-limited parts.
                - Any part for which a mandatory replacement limit is specified in the type design or the maintenance manual.
            - **I**noperative equipment.
                - Refer to the inoperative equipment section below.
            - **C**ompass deviation card.
            - **E**xternal data plate.
                - Located on the left side of the fuselage by the tail.
            - **S**upplemental type certificate.
    - Required equipment:
        - Day VFR required equipment (ATOMATOFLAMES) - 91.205.
            - **A**irspeed Indicator.
            - **T**achometer.
                - For each engine.
            - **O**il Pressure.
            - **M**anifold Pressure.
                - For each altitude engine (turbocharged/supercharged).
            - **A**ltimeter.
            - **T**emperature Gauge.
                - For each liquid cooled engine.
            - **O**il temperature.
                - For every air-cooled engine.
            - **F**uel Quantity Gauge.
                - For each fuel tank.
            - **L**anding Gear Position Lights.
                - If retractable.
            - **A**nti-Collision Lights.
                - Aircraft certified after March 11th, 1996.
            - **M**agnetic direction indicator.
            - **E**LT.

            - Know specifics of 91.207
        - **S**eatbelts and shoulder harnesses.
    - Night VFR required instruments (FLAPS) - 91.205.
        - **F**uses/resettable circuit breakers.
        - **L**anding Light (Electric)
            - for hire operations.
        - **A**nti-Collision Lights.
            - Aircraft certified after August 11th, 1971.
            - Aircraft certified before that date STILL NEED anti-collision lights, but the standards/requirements are different.
        - **P**osition Lights.
        - **S**ource of Electricity (Alternator/Battery).
- Inoperative equipment - 91.213
    - If an aircraft has a Minimum Equipment List, then refer to that first and find out if the aircraft is still airworthy.
    - Determining Airworthiness with inoperative equipment without MEL (TARK):
        - **T**ype certificate data sheet (TCDS).
        - **A**irworthiness Directives.
        - **R**equired by regulations.
        - **K**OEL, Kinds of Operating Equipment List (section 2 of POH).
        - If the instrument or equipment is NOT required:
            - Remove it from the airplane in accordance to 43.9 or deactivate and placard inoperative.

Weather Information

- <u>Quiz the student on weather information.</u>
- Weather briefings:
    - Ask the student to go through the weather briefing they obtained.
    - What type of briefing did they obtain? What are the different types of briefings?
        - Outlook.
        - Standard.
        - Abbreviated.
        - In-flight.
    - Can you get an outlook, then get an abbreviated briefing before the flight and be good to go?
- METARs:
    - Quiz the students on metar valid times.
    - Pull up sample METARs with less common reports on them to see if the student knows or can figure out what is being observed.
- TAF (<u>AC 00-45H</u>):
    - Quiz the student on TAF valid times.

- Pull up sample TAFs with less common reports on them to see if the student knows or can figure out what is being forecast.
- GFA tool:
  - What is the GFA tool?
  - What does it show?
- Winds and temperature aloft chart:
  - How often is it issued?
    - 4 times a day.
  - How long is it valid?
    - Shown on the chart.
  - What does it include?
    - Winds and temps at specific locations.
    - The wind is in true knots.
    - The temperature is given in Celsius.
  - How does it code the winds if they are forecasted to be 100-199 knots?
    - Add 50 to the wind direction and subtract 100 from the speed.
    - Show an example to see if the student can decode it.
  - How does it code the winds if they are forecasted to be calm or less than 5 knots?
    - 9900.
  - How does it code the winds if they are forecasted to be 200 knots or greater?
    - 7799: 270 degrees and greater than 199 knots.
  - Why do some stations not report winds or temperature?
    - No winds are forecasted within 1,500 feet of station elevation.
    - No temperatures are forecasted within 2,500 feet of station elevation.
- Radar imagery:
  - What are the different types of radar?
    - National radar:
      - Shows the intensity of the precipitation.
      - Provides a single view of the entire country.
      - Updated every 10 minutes.
      - Loops the latest 2 hours of images.
    - Single site radar:
      - Composed of 140 radar sites across the U.S.
      - Each site shows base reflectivity, composite reflectivity, and total precipitation.
      - Updated every 10 minutes.
      - Loops the latest 2 hours of images.
- Significant weather prognostic chart:
  - What are the different types? What do they show? How often are they issued?
    - Low level significant weather - surface to FL240.

        - High level - FL250 to FL600.
        - 2 forms:
            - 12 and 24 hour forecast chart:
                - Issued 4 times a day.
                - Valid time printed on chart.
                - Shows forecasted weather to include non-convective turbulence, freezing levels, and IFR or MVFR weather.
                    - Cover the difference between IFR, MVFR, and VFR.
                        - IFR: less than 1000ft ceilings and less than 3SM visibility.
                        - MVFR: 1000ft - 3000ft ceiling and 3-5SM visibility.
                        - VFR: more than 3000ft ceiling and more than 5SM visibility.
            - 36 and 48-hour surface forecast chart:
                - Extension of above chart.
                - Surface weather forecasts.
                - Issued 2 times a day.
                - Pressure patterns, fronts, and precipitation.
- Surface analysis chart:
    - How often are they issued? How long are they valid for? What do they show?
        - Chart depicts the distribution of several items including sea level pressure, the position of the highs, lows, ridges, and troughs, the location and type of fronts, wind direction and speed, temperature and dew point.
            - Analysis of current weather.
            - Computerized.
            - Transmitted every 3 hours.
                - Valid time shown on chart (observed data).
    - What are high- and low-pressure systems?
        - In the northern hemisphere – flow of air from high to low pressure is deflected to the right due to the Coriolis force.
            - High pressure systems (outward - downward - clockwise).
                - Dry, stable, descending air.
                - Good weather.
            - Low pressure systems (inward - upward - counterclockwise).
                - Air flows in a low-pressure area to replace rising air.
                - Bad weather.
    - What is a cold front?
        - Occurs when a mass of cold, dense and stable air advances and replaces a body of warmer air.
        - Moves at 25-30 mph, can move as fast as 60 mph.

        - Rapidly ascending air causes the temperature to decrease suddenly, forcing creation of clouds.
        - It "lifts moist air and creates instability" - recipe for thunderstorms.
        - Prior to cold front passing, cirriform towering cumulus clouds are present, and cumulonimbus clouds may develop.
        - Showery precipitation.
        - After frontal passage, clouds dissipate to cumulus clouds with a corresponding decrease in a precipitation
        - Good visibility prevails and winds from the west and northwest
        - Flight toward a cold front:
            - Flight progresses to the oncoming cold front; the clouds show signs of vertical development with a broken layer.
            - Pilots should use sound judgment on frontal passage.
    - What is a warm front?
        - Occurs when the warm mass of air advances and replaces a body of colder air. Warm fronts move slowly, 10-25 mph. Slope of the advancing front slides over the top of the cooler air and gradually pushes it out of the area. As the warm air is lifted, the temperature drops, and condensation occurs from the humidity of the air.
        - Generally, cirriform, stratiform clouds (stable air) along with fog can be expected to form along the frontal boundary. (rain evaporates when it gets close to the cooler ground).
        - Light to moderate steady precipitation.
        - Flight toward a warm front:
            - Visibility decreases.
            - Falling barometric pressure.
            - Weather overall deteriorates.
            - May be impossible to continue a VFR flight after flying through a warm front.
    - What is a stationary front?
        - When the forces of two air masses are relatively equal, the boundary layer or front that separates them remains stationary and influences the local weather for days.
    - What is an occluded front?
        - Fast moving cold front catches up with a slow-moving warm front. Warm front weather prevails but is followed by cold front weather.
    - What is frontogenesis? What about frontolysis?
        - Frontogenesis: the initial formation of a front.
        - Frontolysis: the dissipation of a front.
- Pilot reports (PIREPs):
    - What are the different types of PIREPs? What is included in a PIREP?

- Weather report from a flight deck. Consists of message type, location, time, altitude/flight level, type aircraft, and at least one other element to describe the report phenomena.
    - 2 types: routine (UA) and urgent (UUA):
        - Urgent includes:
            - Tornadoes.
            - Severe or extreme turbulence.
            - Hail.
            - Low level wind shear within 2000 ft.
    - Contains information regarding conditions as they actually exist.
- When must you submit a PIREP?
    - When requested by another pilot.
    - When the ceiling is at 5,000ft or less.
    - When visibility is 5SM or less.
    - Thunderstorms.
    - Moderate or greater turbulence.
    - Light or greater icing.
    - Wind shear.
    - Volcanic ash.
- Show the student various PIREPs and see if they can decode them properly.

- AIRMETs, SIGMETs, and convective SIGMETs:
    - What is an AIRMET (WA)?
        - Light to moderate weather that may be hazardous to aircraft (primarily light).
            - Issued every 6 hours Valid for 6.
            - Sierra
                - IFR conditions over 50% area. (ceilings less than 1000 ft and vis less than 3sm and mountain obscurations.)
            - Tango
                - Light to moderate Turbulence.
                - surface winds 30 knots or greater.
                - Low-level wind shear.
            - Zulu
                - Moderate icing.
                - Freezing levels.
            - An airmet will be issued if any of these conditions are expected to happen over an area of at least 3,000 square miles.
    - What is a SIGMET (WS)?
        - Severe weather conditions that may be hazardous to aircraft.
            - Unscheduled (comes out when needed).
            - Valid for 4 hours (6 for hurricanes).

            - Severe icing not associated with thunderstorms.
            - Severe or extreme turbulence or CAT not associated with thunderstorms.
            - Dust/sandstorms.
            - Volcanic ash.
        - Issued with alphabetic identifiers (no sierra or tango) → different zones throughout the USA have designated identifiers.
        - Issued when these weather conditions are forecasted to happen over an area of at least 3000 square miles or an area judged to have a significant impact on safety of aircraft.
    - What is a convective SIGMET (WST)?
        - Hazardous convective weather affecting every flight.
            - Issued at 55 minutes past the hour
            - Valid for 2 hours
                - Normal Issuance:
                    - Affects 40% or more of a 3000 sq miles area.
                    - Line of thunderstorms at least 60 miles long with thunderstorms affecting at least 40 percent of its length.
                    - Embedded Thunderstorms
                - Special Issuance:
                    - Tornadoes
                    - Severe thunderstorms with surface winds greater than 50 knots
                    - Hail at surface at least ¾ inch in diameter
            - Numbered sequentially depending on the zone (West, Central, East).
            - If no convective weather, will say "NONE"
- Center weather advisory:
    - What is a center weather advisory? Who issues them? How long are they valid for?
        - A CWA is an aviation weather warning for conditions meeting or approaching national in-flight advisory (AIRMET, SIGMET, or Convective SIGMET) criteria.
        - CWAs are valid for up to 2 hours and may include forecasts of conditions expected to begin within 2 hours of issuance.
- Convective outlook:
    - What is the use of a convective outlook?
        - Consist of both a narrative and a graphic showing severe thunderstorm threats across the US.
        - Convective outlooks are split into 4 different charts:
            - Day 1: risk of severe weather for 24 hours. Issued 5 times a day.
            - Day 2: risk of severe weather for the next 24 hours. Issued 2 times a day.
            - Day 3: risk of severe weather for the next 24 hours. Issued daily.
            - Day 4-8: risk of severe weather for day 4 through day 8. Issued daily.

- AWOS, ASOS, ATIS:
    - What is the difference between AWOS, ASOS, and ATIS?
        - Check the chart supplement to see what type of weather system is in use.
        - AWOS:
            - Automated Weather Observing System.
                - Operated and controlled by the FAA.
                - Automated and reports the weather every minute.
                - Types of AWOS:
                    - AWOS-A: Altimeter setting.
                    - AWOS-AV: Altimeter and visibility.
                    - AWOS-I: Altimeter, wind, temperature, dew point, and density altitude.
                    - AWOS-II: AWOS-I plus visibility.
                    - AWOS-III: AWOS-II plus ceiling.
                    - AWOS-III P: AWOS-III plus precipitation identification.
                    - AWOS-III T: AWOS-III plus thunderstorm/lightning identification.
                    - AWOS-III PT: AWOS-III plus precipitation and thunderstorm/lightning identification.
                    - AWOS-IV: AWOS-III plus precipitation, type and accumulation, freezing rain, thunderstorm, and runway surface sensors.
        - ASOS:
            - Automated Surface Observing System.
                - Operated and controlled by the National Weather Service (NWS), Department of Defense (DoD), and sometimes by the FAA.
                - Used for the national weather system, not just for aviation purposes.
                - Reported information:
                    - Wind speed and direction.
                    - Visibility.
                    - Sky condition.
                    - Ceiling height and precipitation.
                    - Barometric pressure.
                    - Density altitude.
        - ATIS:
            - Automated Terminal Information Service.
                - Towered airports.
                - Requires a human to monitor and broadcast the ATIS.
                - ATIS information is obtained from the ASOS system.
                - Contains more than just weather (NOTAMs, remarks, etc.)

- Published hourly at 55 minutes past the hour.
    - If the weather is rapidly changing, the tower can give an updated ATIS (like a SPECI METAR).
- Reported information:
    - Phonetic alphabet letter.
    - Time of the report.
    - Wind speed and direction.
    - Visibility.
    - Cloud cover.
    - Temperature.
    - Dew point.
    - Barometric pressure.
    - NOTAMs, remarks, other information.

- Thunderstorms:
    - What you need for a thunderstorm:
        - Lifting action.
        - High moisture.
        - Unstable air.
    - Stages:
        - Developing / Cumulus:
            - towering cumulus (rising air), little rain, occasional lightning.
        - Mature:
            - fully formed, heavy rain, frequent lightning, strong winds.
            - Starts when rain makes contact with the surface.
        - Dissipating:
            - downdrafts, rainfall decreases, still producing strong winds and lightning.
    - Types:
        - Single cell: develop on warm humid days. severe, produce hail, and microburst.
        - Multi-cell: develop in clusters, cover large areas, can move in different directions.
        - Squall line: a narrow band of TS, ahead of a cold front (moist, unstable air, lifting action). Severe and wide.
        - Supercell: Single long-lived TS. large hail and significant tornadoes.
        - Steady State
        - Air-mass
    - What to expect:
        - Lightning.
        - Turbulence.
        - Hail.
        - Tornadoes.
        - Wind shear.

        - Icing.
- Turbulence
    - Types?
    - Intensities?
    - Frequencies?
- Icing:
    - Intensities:
        - Trace
        - Light
        - Moderate
        - Severe
    - Performance impacts:
        - Drag → increase.
        - lift → decrease.
        - Weight → increase.
        - Stall speed → increase.
    - Icing Locations
        - Structural:
            - Requires visible moisture and aircraft surface below freezing.
            - Clear
            - Rime
            - Mix
            - Frost
        - Induction icing:
            - Carb ice → can happen with temperatures as high as 100F/38C and a relative humidity of 80%.
                - Air in the carburetor is accelerated in the venturi, causing the temperature of the air to decrease. More susceptible to icing.
                - Turn carb heat on (small reduction in RPM, warm air is less dense than cold air, enriches the mixture).
        - Instrument icing:
            - Pitot tube and static ports.
            - Static wicks.
            - Antennas.
            - Stall warning tab.
            - OAT probe.
- Wind shear:
    - Dramatic change in wind speed and/or direction.
    - Low-level.
        - Result of temperature inversion.

        - Below 2000 feet AGL.
    - Close isobars on charts indicate a strong pressure gradient which results in strong winds.
    - Winds flow across isobars at the surface due to friction and reduction of Coriolis force.
    - Winds flow parallel to the isobars at altitude (Winds aloft).
- Microburst:
    - Occurs in a space of less than one mile horizontally and within 1000 feet vertically.
    - Can span for 15 min.
    - Produce downdrafts of 6,000 feet per minute.
    - Can produce hazardous wind direction change of 45 degrees or more in a matter of seconds.
    - The plane will encounter an increase in headwind, then performance decreasing downdrafts, then a shear tailwind, and can result in terrain impact.
    - It happens in confined areas.
- Sea breeze and land breeze:
    - Land warms faster than water – air over land becomes warmer and less dense. It rises and is replaced by cooler, denser air flowing over the water.
        - This causes onshore wind called sea breeze.
    - At night – land cools faster than water, as does corresponding air. Warmer air over the water rises and is replaced by cooler, dense air from the land.
        - This causes offshore wind called land breeze.
- Stability of the atmosphere:
    - Stability of the atmosphere depends on its ability to resist vertical motion.
    - Rising air expands and cools due to the decrease in air pressure as altitude increases.
    - Opposite is true of descending air.
    - The combination of moisture (makes air lighter and enables air to rise) and temperature (high temperature leads to lifting action (air is expanding, less dense, lighter)) determine the stability of the air and the resulting weather.
        - LIFTED K INDEX https://www.weathertap.com/guides/aviation/lifted-index-and-k-index-discussion.html
        - LAPSE RATE
        - http://www.meteo.psu.edu/wjs1/Meteo3/Html/stability.htm
- Inversions:
    - An inversion is when temperature increases as altitude increases.
    - Surface-based temperature inversions occur on clear, cool nights when the air close to the ground is cooled by the lowering temperature of the ground.
        - The air within a few hundred feet of the surface becomes cooler than the air above it.
        - Frontal inversions occur when warm air spreads over a layer of cooler air.
- Fog:
    - Temperature of air near the ground is cooled to the air's dew point.
    - Types (reported when visibility is less than 5/8th of a mile):

        - Radiation fog – Occurs when the ground cools rapidly due to terrestrial radiation, and surrounding air reaches dew point. As the sun rises and temperature increases, the fog burns off.
        - Advection fog – when a layer of warm moist air moves over a cold surface. A 15-knot wind is necessary. Common to coastal areas where sea breezes can blow the air over cooler landmasses.
        - Upslope fog – Occurs when moist, stable air is forced up sloping land features like a mountain range. Also requires wind for formation and continued existence.
        - Steam fog – forms when cold dry air moves over warm water. As water evaporates it turns into smoke.
        - Precipitation fog – Occurs when the raindrops are warmer than the surrounding air.
        - Ice Fog

Cross-Country Flight Planning

- <u>Quiz the student on cross-country flight planning.</u>
- <u>Inspect the nav log, sectional chart, and TOLD card before leading into scenarios.</u>
- <u>Ensure all boxes are filled in correctly and appropriately.</u>
- <u>Ensure that the TOLD card fuel calculations match what is on the nav log.</u>
- Airport information:
    - Have the student brief information from the chart supplement:
        - Frequencies.
        - Runway lengths.
        - Fuel services.
        - Any LAHSO information.
        - Airport lighting.
        - Remarks.
- Navigation log:
    - Make sure the flight plan form is filled out correctly.
    - Ask the student why the cruising altitude was chosen - 91.159.
    - Ask the student why it is a safe altitude - 91.119.
    - Evaluate fuel, time, and distance totals.
    - Ask if TAS increases with wind.
    - Discuss the effects of density altitude on TAS.
    - Ask the student what information was required to plan this flight - 91.103.
    - Discuss cruising performance values:
        - Top of climb.
        - Fuel requirement – discuss mixture leaning techniques.
            - Best economy vs best power.
        - Engine RPM.
        - True airspeed.
        - Top of descent.

        - Takeoff/landing data.
        - Range and endurance, what's the difference?
- NOTAMs:
    - Ensure that the student has the applicable NOTAMs on the nav log.
    - Quiz the student on the different types of NOTAMs.
        - Types (AIM 5-1-3):
            - NOTAM (D).
            - FDC NOTAM.
            - International NOTAM.
            - Military NOTAM.
- Pilotage and dead reckoning:
    - What is pilotage? What is dead reckoning?
        - Pilotage: navigation by reference to landmarks or checkpoints.
        - Dead reckoning: navigation solely by computation based on time, airspeed, distance, and direction.

National Airspace System

- Check for airspace recognition on the sectional chart.
- Quiz the student on chart symbology.
- Ask the student about the types of airspace
    - Discuss weather minimums where appropriate.

- Controlled and uncontrolled airspace - 91.127 through 91.135:
    - Class A:
        - What is class A airspace?
            - Generally, from FL180 to FL600.
            - Extends 12 NM offshore of 48 contiguous states and Alaska.
        - What are the entry requirements and required equipment?
            - Entry requirements:
                - ATC Clearance.
                - IFR flight plan.
            - Required equipment:
                - Two-way radios.
                - Mode C transponder.
                - ADS B-OUT.
        - What are the weather minimums?
            - None, IFR.
    - Class B:
        - What is class B airspace?
            - Generally, from SFC to 10,000 ft MSL.
            - Busy airports: based on total operations or passenger enplanements.

        - Multiple layers (upside-down cake).
        - Configuration can be different depending on the airport.
        - ATC provides separation within the airspace.
        - Some airports do not allow student, sport, or recreational pilots to land, usually due to high traffic volume
            - Part 91, appendix D, section 04.
    - What are the entry requirements and required equipment?
        - Entry requirements:
            - ATC Clearance.
        - Required equipment:
            - Two-way radios.
            - Mode C transponder (in class B, above and within the lateral limits of class B, and within 30 nm (mode C veil)).
            - ADSB-OUT (in class B, above and within the lateral limits of class B, and within 30 nm (mode C veil)).
    - What are the weather minimums?
        - 3SM visibility and clear of clouds.
- Class C:
    - What is class C airspace?
        - Generally, from SFC to 4,000 ft MSL.
        - Surrounds airports that have a control tower, radar approach control, and have a certain number of IFR operations or passenger enplanements.
        - Configuration can be different depending on the airport.
        - Two layers:
            - Inner ring extends from the SFC to 4,000 ft (typically) with a 5 NM radius.
            - Outer ring extends from 1,200 ft to 4000 ft (typically) with a 10 NM radius.
    - What are the entry requirements and required equipment?
        - Entry requirements:
            - Establish two-way radio communication.
        - Required equipment:
            - Two-way radios.
            - Mode C transponder (in class C and above and within the lateral limits of class C)
            - ADSB-OUT (in class C and above and within the lateral limits of class C).
    - What are the weather minimums?
        - 3SM visibility and 1,000 ft above, 500 ft below, 2000ft horizontally.
- Class D:

        - What is class D airspace?
            - Generally, from SFC to 2,500 ft MSL.
            - Surrounds airports that have a control tower.
            - Configuration can be different depending on the airport (KMLB!)
            - Usually designed to contain instrument approaches.
        - What are the entry requirements and required equipment?
            - Entry requirements:
                - Establish two-way radio communication.
            - Required equipment:
                - Two-way radios.
        - What are the weather minimums?
            - 3SM visibility and 1,000 ft above, 500 ft below, 2000ft horizontally.
    - Class E:
        - What is class E airspace?
            - Controlled airspace that is not class A, B, C, or D.
            - Extends up from either the surface or a designated altitude to the overlying or adjacent controlled airspace.
            - Sectional and other charts depict all locations of Class E airspace with bases below 14,500 feet MSL. In areas where charts do not depict a class E base, class E begins at 14,500 feet MSL.
            - In most areas, the Class E airspace base is 1,200 feet AGL. In many other areas, the Class E airspace base is either the surface or 700 feet AGL.
            - All airspace above FL 600 is Class E airspace.
        - What are the entry requirements and required equipment?
            - ADSB-OUT: at or above 10,000 feet MSL, excluding airspace at and below 2,500 feet AGL; over the Gulf of Mexico, at and above 3,000 feet MSL, within 12 nm of the U.S. coast.
            - Mode C transponder: at and above 10,000 feet MSL, excluding the airspace at and below 2,500 ft AGL.
        - What are the weather minimums?
            - Less than 10,000ft MSL: 3SM visibility and 1,000 ft above, 500 ft below, 2000ft horizontally.
            - More than 10,000ft MSL: 5SM visibility and 1,000ft above, 1,000ft below, 1SM horizontally.
    - Class G:
        - What is class G airspace?
            - Uncontrolled.
            - ATC has no authority or responsibility.
                - Although ATC has no authority, a pilot cannot be careless or reckless as this would break the 91.13 regulation.

                - VFR rules apply.
            - What are the entry requirements and required equipment?
                - None.
            - What are the weather minimums?
                - Less than 1,200ft AGL:
                    - Day: 1SM visibility and clear of clouds.
                    - Night: 3SM visibility and 1,000 ft above, 500 ft below, 2000ft horizontally.
                - More than 1,200ft AGL and less than 10,000ft MSL:
                    - Day: 1SM visibility and 1,000 ft above, 500 ft below, 2000ft horizontally.
                    - Night: 3SM visibility and 1,000 ft above, 500 ft below, 2000ft horizontally.
                - More than 10,000ft MSL:
                    - 5SM visibility and 1,000ft above, 1,000ft below, 1SM horizontally.
        - Special use airspace:
            - What are the different types of special use airspace?
                - **W**arning area.
                    - The government does not have jurisdiction over airspace.
                    - Defined dimensions, extending 3NM offshore of the U.S.
                    - Contains activity that may be hazardous to nonparticipating aircraft.
                    - Charted as "W" followed by a number.
                - **A**lert area.
                    - High volume of pilot training or an unusual type of aerial activity.
                    - Depicted with an "A" followed by a number on charts.
                - **R**estricted area.
                    - Area in which operations are hazardous to non-participating aircraft.
                        - Artillery firing, guided missiles.
                    - Flight of aircraft may be restricted, but not prohibited.
                    - Must have permission from ATC or the controlling agency to enter.
                    - Charted with an "R" followed by a number.
                    - Information is found on the back of charts.
                - **M**ilitary operation area.
                    - Defined vertical and lateral limits.
                    - Separates certain military training activities from IFR traffic.
                    - IFR traffic may be cleared through airspace if ATC can provide separation.
                    - Additional information for each is found on the back of the sectional.
                - **P**rohibited area.
                    - Airspace in which flight is prohibited.

            - Camp David, White House, Congressional buildings.
        - Usually for security or national welfare.
        - Charted at “P” followed by a number.
    - Controlled firing area.
        - Contains activity that is hazardous to non-participating aircraft.
        - Activity must stop if the spotter spots non-participating aircraft.
        - Published in chart supplement under Special Notices.
- Other airspace:
    - What are some other types of airspace?
        - Military Training Routes (MTRs):
            - Used by military aircraft to maintain proficiency in tactical flying.
            - Usually established below 10,000 feet MSL for operations exceeding 250 knots.
            - Routes depicted as IR (if IFR) and VR (if VFR), followed by a number:
                - Those not exceeding 1,500 feet AGL contain 4 numbers (IR 1206).
                - Those above 1,500 feet AGL contain 3 numbers (VR 207).
        - Parachute jump aircraft operations:
            - Airports at which there are high numbers of parachute operations.
            - Published in chart supplement unde Parachute Jumping Areas and depicted on sectional charts.
        - Published VFR Routes
            - Used for transitioning around, under, or through complex airspace
            - Usually found on VFR terminal area planning charts
            - Often referred to as VFR flyway, VFR corridor, Class B airspace VFR transition route, and terminal area VFR route
        - Terminal Radar Service Areas (TRSAs)
            - Areas in which pilots can receive additional radar services.
            - Purpose is to provide separation for IFR and participating VFR traffic.
            - The airspace within the TRSA becomes Class D around the airports.
            - Remaining portion overlies other controlled airspace, which is normally Class E.
            - Depicted on VFR sectional charts and terminal area charts with a solid black line and altitudes for each segment.
            - Participation is voluntary and encouraged.
        - Temporary flight restrictions (TFRs)
            - Designated by flight data center (FDC) Notice to Airmen (NOTAM)
                - Begins with phrase “FLIGHT RESTRICTIONS” followed by location, effective time period, and defined in statute miles, and altitudes affected.

        - Purposes may include:
            - Protect persons or property in air or on surface from existing or imminent hazard.
            - Provide a safe environment for operation of disaster relief aircraft.
            - Prevent unsafe congestion of sightseeing aircraft above incident or event.
            - Protect declared national disasters for humanitarian reasons in Hawaii.
            - Protect the President, Vice President, or other public figures.
            - Provide a safe environment for space agency operations.
        - If penetrated, the pilot may undergo security investigations and certificate suspensions.

Performance and Limitations

- TOLD card:
    - Have the student take out the TOLD card that was to be prepared for this lesson.
    - Quiz the student on the following:
        - Weight and balance.
            - Arm, datum, moment, taxi fuel burn, and trip fuel burn.
            - Effects of forward and aft CG.
            - Effects of weight on performance.
        - Pressure altitude and density altitude.
            - What do they mean?
            - Why is it important?
        - Performance charts.
            - Ask them to show you how they got the takeoff and landing distances.
        - Aircraft speeds.
            - Ensure that the student can explain all of the different V speeds.
- Atmosphere:
    - What is standard atmosphere?
        - 29.92” Hg (mercury) – decreases 1” per 1,000ft.
        - 15 degrees Celsius – decreases 2 degrees per 1,000ft.
    - How do temperature, density, humidity, and pressure affect performance?
        - Lower temperature = higher density = better performance.
        - Lower humidity = heavier air = higher density = better performance.
        - Higher pressure = higher density = better performance.

Operation of Systems

- Quiz the student on aircraft systems:
    - Aircraft structure.

        - What is the material of the aircraft skin?
        - What kind of structure does the aircraft have?
        - What is the aircraft structure composed of?
    - Primary and secondary flight controls.
        - What are the primary flight controls?
            - What type of ailerons do we have?
            - Why do we have differential ailerons?
            - What is adverse yaw?
            - What is the difference between a stabilator and an elevator?
            - What is an anti-servo tab and what purpose does it have?
        - What are the secondary flight controls?
            - What type of flaps do we have?
            - What purpose do they serve?
            - Why is there a slot between the flap and trailing edge of the wing?
            - How does trim work?
            - Do we only have stabilator trim?
    - Powerplant.
        - Tell me what you know about the powerplant.
            - Students should cover the CL4HAND acronym.
            - Students should know what O-320-D3G means.
    - Propeller.
        - Students should know the manufacturer and length of the propeller.
        - Turning tendencies:
            - Torque (roll):
                - Involves Newton's 3rd law – for every action there is an equal and opposite reaction.
                    - Engine parts and propeller move one way (right), aircraft wants to move the other way (left).
            - Spiraling slipstream (yaw):
                - Spiraling rotation of slipstream from rotation of prop
                - Strikes vertical stabilizer causing rotation around the vertical axis (left yaw).
            - P-factor (yaw):
                - P-Factor → high angle of attack and high power.
                    - Downward blade produces more thrust because of the increased angle of attack between relative airflow and chord line of propeller.
                    - This results in the airplane yawing to the left.
            - Gyroscopic precession:
                - When a force is applied, the resulting force takes effect 90 degrees

ahead of and in the direction of rotation.
    - When pitching up, the force is applied to the bottom of the propeller. Since it spins clockwise the force is felt 90 degrees in the direction of rotation causing a right yawing tendency.
- Oil system.
    - What are the types of oil and when they are used?
    - What are the required oil quantities? (Lycoming and FIT)
    - What is the difference between wet-type and dry-type oil sumps? What does our aircraft have?
    - What are the functions of the oil?
- Landing gear system.
    - What type of landing gear system does the Warrior have?
    - What type of struts does the Warrior have?
    - How much should the main struts be extended? What about the nose strut?
    - How inflated should the nose and main tires be?
    - What do you check during the pre-flight?
- Hydraulic system.
    - What does the hydraulic system do?
    - How many cylinders does the system have? How about reservoirs?
    - What type of hydraulic fluid does the system use?
- Fuel system.
    - Ask the student to draw the fuel system on the board.
    - How many gallons of fuel can the Warrior hold? Usable?
    - What's the purpose of the gascelator?
    - Why does the system have two fuel pumps?
    - Why does the primer only prime 3 cylinders?
    - How does the carburetor work?
- Pitot-static system.
    - Ask the student to draw the pitot-static system on the board.
    - How does the airspeed indicator work? (briefly)
    - How does the altimeter work? (briefly)
    - How does the VSI work? (briefly)
    - What systems does the Warrior have in case of a blocked static/pitot?
- Vacuum system.
    - Ask the student to draw the vacuum system?
    - What instruments are powered by the vacuum system?
    - What could cause a vacuum system malfunction?
    - What indications do you have that would tell you the vacuum system has failed?
- Electrical system.
    - What are the components of the electrical system?

        - How does the alternator work?
        - How many amps and volts does the alternator have?
        - How many amps and volts does the battery have?
        - Why does the alternator have 2 more volts than the battery?
        - What does the voltage regulator do? What about the overvoltage relay?
        - How many times can you reset a circuit breaker?
        - Can the engine start if the battery dies?

Aeromedical Factors

- What is hypoxia?
    - Lack of oxygen.
- What are the 4 types?
    - Hypoxic - insufficient oxygen available to the body.
        - Can be caused by a blocked airway.
        - Not the same amount of oxygen at higher altitude – same ratio, but just less air molecules.
    - Hypemic - blood is unable to transport oxygen.
        - Inability to absorb oxygen.
        - Caused by CO poisoning, blood loss, anemia.
    - Stagnant - blood is not flowing to move oxygen.
        - Inability to transport the oxygen.
        - Caused by a limb falling asleep.
    - Histotoxic - inability of cells to use oxygen.
        - Caused by usage of drugs and alcohol.
    - Symptoms of hypoxia (these will differ from person to person!):
        - Cyanosis (blue nails and lips).
        - Headache.
        - Impaired judgment.
        - Decreased reaction time.
        - Euphoria.
        - Visual impairment.
        - Drowsiness.
        - Lightheadedness or dizziness.
        - Tingling in fingers or toes.
        - Numbness.
    - Corrective actions:
        - Use supplemental oxygen (requirements in 91.211).
        - Decrease altitude.
        - Use medication.
        - Don't drink and fly.
        - Stretching – G suits.

- What is hyperventilation?
    - Excessive rate and depth of respiration that leads to the abnormal loss of carbon dioxide from the blood or too much oxygen. Caused by anxiety, fear, or stressful situations.
        - Symptoms (can be similar to hypoxia):
            - Increased breathing rate.
            - Unconsciousness.
            - Visual impairment.
            - Lightheadedness or dizziness.
            - Tingling feelings.
            - Hot/cold sensations.
            - Muscle spasms.
        - Corrective actions:
            - Breathing normally.
            - Breath into paper bag.
            - Talking aloud.
- What causes middle ear and sinus problems?
    - Gasses trapped within the body expand with altitude causing pain in the middle ear and sinuses if gasses are not released. Usually caused due to change in pressure from climbing and descending with blockage.
        - Causes:
            - Illness and blockages.
        - Ear:
            - Pressure builds inside the middle ear – needs to be relieved when at altitude or when back on the ground.
            - Symptoms:
                - Can have loss of hearing.
                - Eardrum could rupture.
                - Infections.
            - Corrective actions:
                - Valsalva maneuver.
                - Yawning.
                - Swallowing.
        - Sinuses (most frequently experienced during descent):
            - Symptoms:
                - Pain in sinuses.
                - Toothache.
                - Bloody mucus discharge.
            - Corrective actions:
                - Slow descent.
                - Don’t fly.

        - The Eustachian tube connects your throat with your ear – can cause you to be sick.
- What is spatial disorientation?
    - Lack of orientation in regard to position, altitude, or movement of the airplane.
    - 3 systems:
        - Vestibular – organs in the inner ear that sense position by the way they are balanced.
            - 3 semicircular canals (based on 3 axes of motion) with fluid and tiny hairs that sense which way fluid is moving in the ear:
                - Horizontal.
                - Superior.
                - Posterior.
            - Illusions:
                - The leans – Once in a constant bank, (usually for 30 seconds), the vestibular system will "catch up" with the plane and the pilot will sense that the aircraft is straight and level.
                - Coriolis illusion – after banking for a while and making a sudden head movement, it can give the feeling of a maneuver or turn not occurring. Pilots might make wrong corrections and make things worse.
                - Graveyard spiral – prolonged coordinated turn can feel straight and level, could be due to the leans, (in a turn speed decreases and vertical lift decreases). The pilot will sense the altitude loss and add elevator pressure. But because the aircraft is actually banked, the turn will steepen, and the nose will drop more. Leads to an uncontrolled spiral dive.
                - Graveyard spin – enter a spin which is maintained and felt to decrease, so the opposite rudder is applied and then it feels like a spin in the opposite direction which will be corrected by use of rudder and then a spin in the original direction.
        - Somatosensory – nerves in the skin, muscles, and joints, which, along with hearing, sense position based on gravity, feeling and sound.
            - Illusions:
                - Somatogravic illusion – rapid acceleration gives the feeling of being nose-up. Pilots may push the nose down unnecessarily.
                - Inversion illusion – abrupt change from climb to straight and level can give the feeling of tumbling backwards. Pilot may push the aircraft nose lower.
                - Elevator illusion – abrupt vertical acceleration may give false feeling of being in climb causing pilot to push nose lower.
        - Visual – eyes, which sense position based on what is seen.
            - Illusions:
                - False horizon – clouds or other obscured horizons can give the illusion

        of a false horizon. This can lead to the pilot using the wrong horizon reference line.
    - Runway width - A narrower-than-usual runway can create an illusion that the aircraft is at a higher altitude than it actually is. This can lead to the pilot flying a lower approach or rounding out too late. A wider runway can have the opposite effect with the risk of the pilot leveling out the aircraft high and landing hard or overshooting the runway.
    - Runway slope - an upsloping runway can create an illusion that the airplane is higher than it actually is. This can lead to the pilot flying a lower approach. A down sloping runway can create an illusion that the airplane is lower than it actually is. This can lead to the pilot flying a higher than normal approach.
    - Featureless terrain - An absence of ground features (like over the water, dark areas, desert, etc.) can create an illusion that the airplane is higher than it actually is. This can lead to the pilot flying lower or a lower approach.
- What is motion sickness?
  - Caused by brain receiving conflicting messages about the state of the body.
    - Symptoms:
      - Loss of appetite.
      - Dry mouth.
      - Sweating.
      - Nausea.
      - Headaches.
    - Corrective actions:
      - Open air vents.
      - Focus on objects outside of airplane and in the distance.
      - Avoid unnecessary head movements.
- What is carbon monoxide poisoning?
  - Colorless, tasteless, odorless gas that prevents hemoglobin from carrying oxygen to the cells.
    - Causes:
      - Inhalation of combustion fumes.
      - Cars, planes, charcoal grills.
    - Symptoms:
      - Headache.
      - Drowsiness.
      - Dizziness.
    - Corrective actions:
      - Shut off heater.

            - Open air vents.
            - Seek medical attention.
- Fatigue and Stress:
    - Fatigue:
        - Acute – felt after long periods of physical strain.
            - Short term.
            - Coordination and alertness may be reduced.
            - Prevented by adequate sleep, exercise, nutrition.
        - Chronic fatigue - occurs by not recovering from acute fatigue.
            - Long term.
            - Performance can decrease.
            - Prolonged periods of rest are needed.
    - Stress:
        - Body's response to demands placed on it.
        - Physical and psychological.
        - Small levels are ok.
            - Can be a burst of energy/motivation.
        - Avoid by exercising, well nourished, and well rested.
- Dehydration and heatstroke – lack of fluids for the body to carry for normal functions.
    - Dehydration – occurs by inadequate intake, or loss of, fluids.
        - Drink plenty of fluids.
        - Avoid caffeine and alcohol intake.
        - Symptoms:
            - Headaches.
            - Nausea.
            - Internal organs can shut down.
            - Can lead to heatstroke.
    - Heatstroke – body is unable to keep itself cool.
        - Similar symptoms to dehydration but can lead to complete collapse.
- Effects of alcohol and drugs – FAR 91.17:
    - Never combine drugs, alcohol, and flying.
    - 1 oz. of liquor, 1 bottle of beer, or 4oz. of wine can impair flying abilities.
    - Maximum .04 BAC.
    - Can increased effects of hypoxia and dehydration.
    - FAA prohibits cockpit operations if you consumed alcohol within 8 hours (12 hours for FIT).
    - If medication must be taken, ensure it is approved by FAA (ask AME).
- Effects of nitrogen excess during scuba dives:
    - Nitrogen is absorbed into blood stream during dives due to increased pressure and can expand and cause severe pain.
        - After uncontrolled ascent:

            - Wait at least 12 hours to fly up to 8,000 ft. after diving.
            - Wait at least 24 hours before flights above 8,000 ft.
        - After controlled ascent:
            - Wait 24 hours before flying at any altitude.
- Types of decompression sickness – nitrogen bubbles.
    - Bends – joints, elbows, shoulders, lips.
    - Chokes – in the lungs.
    - Skin – itching on the skin, swelling and skin compressions.
    - Neurologic – Brain, spinal cord, and peripheral nerves.
    - Recovery techniques:
        - Land as soon as possible.
            - Declare emergency if needed.
- Barometric chamber if needed.

Radio Communications and ATC Light Gun Signals

- Give the student a scenario where they have a radio communication failure in the practice area.
- Make sure they attempt to troubleshoot.
- If they decide to proceed to KMLB, quiz them on how they would enter the airspace/pattern and what light gun signals they are looking for.

Color and Type of Signal	Aircraft on the Ground	Aircraft in Flight
Steady Green	Cleared for takeoff	Cleared to land
Flashing Green	Cleared for taxi	Return for landing
Steady Red	STOP	Give way and continue circling
Flashing Red	Taxi clear of runway in use	Airport unsafe, do not land
Flashing White	Return to starting point on airport	----
Alternating Red and Green	Exercise extreme caution	

References: 14 CFR 91.125; AIM 4-3-13

Airport, Runway, and Taxiway Signs, Marking and Lighting

- Quiz the student on various airport, runway, and taxiway signs, markings, and lighting.
- https://www.aopa.org/-/media/files/aopa/home/online-education/flash-cards/rwcards_lo.pdf

Night Preparation

- Definitions of nighttime:
    - FAA definition (FAR Part 1 - Definition and Abbreviations) - logging nighttime: end of evening civil twilight to the beginning of morning civil twilight.
    - Position lights (FAR Part 91 - 91.209): must be on between sunset and sunrise.
    - Nighttime passenger currency (FAR Part 61 - 61.57): currency can be accomplished between 1 hour after sunset and 1 hour before sunrise. The minimum of 3 landings must

be to a full stop.

- Notes:
    - Nighttime hours can be found in the Airman's Almanac.
    - Alaska requires position lights "during the period a prominent unlighted object cannot be seen from a distance of 3 statute miles or the sun is more than 6 degrees below the horizon" (91.209).

- Night vision:
    - How do the eyes work?
        - Rods and cones:
            - Two types of light nerve endings which transmit messages to the brain via the optic nerve.
                - Cones – Responsible for color, detail, and far away objects.
                    - The cones are located in the center of the retina.
                    - The fovea is where almost all of the light-sensing cells are cones.
                        - This is the center of vision.
                        - Need light in order to function.
                - Rods – Responsible for peripheral vision and vision in dim light.
                    - The rods are located around the cones (peripherals).
                    - Scan using off center viewing at night.
                    - When attempting to find traffic do not stare directly at it, look slightly off to the left or right to allow the rods to see the aircraft.
                    - The rods can take approximately 30 minutes to fully adapt to the dark.
                        - Therefore, it is important to avoid bright lights before and during flight.
                            - This is why red flashlights are recommended during flight, they do not disrupt the rod's dark adaptation as much.
        - Due to the increase in the body's need for oxygen at night (increased eye muscle usage, concentration, etc.), the AIM recommends using supplemental oxygen at or above 5,000ft.
            - AIM 8-1-2.
- Night illusions:
    - Autokinesis:
        - Caused by staring at a single point of light against a dark background creating the illusion that the light appears to move.
        - Prevent this by focusing the eyes on objects at varying distances and avoid fixating.
            - Keep the eyes moving and offset/use peripherals.
    - False horizon:

        - Caused when the natural horizon is obscured/not readily apparent (night sky and ocean)
            - Generated by confusing bright stars and city lights.
        - Use and trust your instruments to maintain orientation.
    - Featureless terrain / black hole approach:
        - An absence of ground features can create the illusion that the aircraft is higher than it actually is.
        - This results in a tendency to fly a lower than normal approach.
        - Use and trust your instruments to maintain orientation.
    - Ground lighting:
        - Regularly spaced lights along a road/highway etc. can appear to be runway lights.
        - Bright runway or approach lights can create the illusion the airplane is closer to the runway
        - Mitigate this as much as possible by maintaining situational awareness.
            - Know what to expect to see (type of airport/runway lighting), where you expect to see it and know where you are (use navaids, GPS, landmarks, etc).
- Lighting:
    - Aircraft lights:
        - Far Part 91.205 requires the airplane to be equipped with the following lights at night:
            - Landing light (electric) - if the airplane is for hire.
            - Anti-collision lights.
            - Position lights.
    - Airport lighting (AIM 2-1):
        - Taxiway lighting:
            - Edge lights - blue.
            - Centerline lights - green, alternates green/yellow on the runway (lead off lights).
            - Runway guard lights - yellow.
            - Taxiway clearance bar lights - yellow.
        - Approach lighting:
            - VASI/PAPI: red and white.
            - Approach lights: ALSF-2, ALSF-1, SSALR, MALSR, MALSF, ODALS.
            - Refer to chart supplement to see what approach lighting system and glidepath indicator an airport is equipped with.
        - Runway lighting:
            - Entry threshold - green.
            - Runway edge lights - white.
                - Last 2,000ft - amber.
            - Centerline lights - white.

            - Last 3,000ft - red and white.
            - Last 1,000ft - red.
        - Runway touchdown zone lights - white.
        - Runway end identifier lights - red.
- Preparation and required equipment:
    - Preflight inspection - 91.103:
        - Required equipment for VFR flight at night:
            - ATOMATO FLAMES and FLAPS – 91.205 Parts B and C.
                - Fuses (if applicable).
                    - Needed – 1 set of spare fuses or 3 spare fuses for each kind required.
                - Landing Light (electric) - for hire.
                - Anti-collision light.
                - Position Lights.
                - Source of Power.
        - Pilot equipment:
            - Flashlight: red and white.
                - White light is used to preflight the airplane while the red light is used while in flight.
                    - Again, red light will not impair night vision.
        - Walk around:
            - The preflight inspection is still necessary.
            - White light flashlight is good for the inspection (red light inside the cockpit).
            - Check all aircraft lights.
            - Check the ramp for obstructions.
            - Check NOTAMs, weather, obstructions, navigation systems and radios.
            - Fuel minimums at night are 45 minutes (FAA) and 60 minutes (FIT).
        - In-flight orientation:
            - Relying on instruments at night is needed due to the absence of visual references.
            - Be familiar with runway/airport lights: know what you are looking for!
            - Use the HSI to bug in the runway heading.
            - Don't be afraid to ask ATC for help.
            - Pilot Controlled Lighting:
                - High Intensity Runway Lighting – 7 clicks.
                - Medium Intensity Runway Lighting – 5 clicks.
                - Low Intensity Runway Lighting – 3 clicks.
            - Round out and flare:
                - Judgment of height, speed, and sink rate may be impaired.
                    - There is often a tendency to round out too high.

        - A good rule is to start the round out when the landing lights reflect on the tire marks on the runway.
- Visual scanning and collision avoidance:
    - Emphasize that it is the pilot in command's responsibility to avoid traffic.
    - IF YOU SEE SOMETHING, SAY SOMETHING!
    - At night, the use of peripheral vision is the best method to scan for traffic.
    - Use the same scanning method but paying more attention to your peripheral vision. If a plane is spotted, avoid looking directly at it. Look slightly to the side of it.

Wake Turbulence Avoidance
- What is wake turbulence?
        - Pressure differential → low pressure above and high pressure below the wing.
        - Air moves from the high-pressure area under the wing to the low-pressure area over the wing.
            - It goes around the wing tip and then trails behind.
            - As air curls around the tip, it combines with the wash to form a fast spinning trailing vortex.
            - Right wing tip - counterclockwise.
            - Left wing tip - clockwise.
- How can we avoid wake turbulence?
    - Explain how to avoid wake turbulence:
        - Vortices are greatest when an aircraft is in a heavy, slow, and clean configuration.
            - Avoid flying through another aircraft's flightpath.
            - Rotate prior to proceeding aircraft.
            - Delay takeoff by 2-3 minutes.
            - Avoid following aircraft on similar flight paths.
            - Land past the touchdown point of the other aircraft.
        - Wind is an important factor to wake turbulence.

Low Level Windshear
- This was already covered in the weather section above.

Charts and Publications
- This was already covered in the cross-country planning section above.

Emergency Equipment and Survival Gear
- Fire extinguisher:
    - Check in preflight to make sure it is properly charged (in the green), the pin is in, and it is latched.
    - Operation: open the latch, pull the pin out, point at the base of the fire and spray. If used in flight, make sure the vents and windows are opened.
- Survival kit:

- A survival kit should provide sustenance, shelter, medical care and a means to summon help (radio, phone). Consider the terrain, the climate, and type of emergency communication equipment needed:
  - Winter: include blankets, extra clothes and jackets if available
  - Summer: blankets might be a good idea because it can still get cool at night
  - Mountainous terrain in December requires different survival gear than ocean flying in August.

- Overwater operations:
  - 91.509.
- Emergency locator transmitter (ELT):
  - These are designed to transmit a signal on 121.5 at a frequency of 243 megahertz (406 on newer ELTs).
    - 121.5: sounds like a loud beeping noise. The closer you are, the louder the sound.
    - 243/406 MHz: sends a signal containing the aircraft's location, pilot's name, contact information, and type of aircraft.
  - Usually armed to go off at an impact of around 5G's. Can also be manually turned on in the cockpit (ARM vs ON).
  - Batteries must be replaced after one hour of cumulative use or half the useful life is up.
  - Inspection is every 12 calendar months.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 143.
- Assign homework:
  - Review all of the stage 1 and stage 2 maneuvers to be performed for the solo.

Completion Standards

Through oral questioning the instructor shall determine that all knowledge areas meet the Private Pilot Airman Certification Standards.